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Introducing the Excel Tariff Calculator

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Feature Overview

- Classical actuarial methodology based on commutation functions and present values
- Simple worksheet for calculating premiums and projection values
- Additional worksheet with the mortality tables
- VBA modules for constants, commutation functions, and present values (in total around 400 lines of code)

Worksheets for Calculation and Mortality Table

	A	B	C	D	E	F	G	H	I	J	K	L
1	**Endowment Life Insurance Calculator**									VBA solution with caching commutation values		
2												
3	**Policy data**			**Tariff data**			**Limits**			**Premium calculation**		
4	x	40		InterestRate	1,75%		MinAgeFlex	60				
5	Sex	M		MortalityTable	DAV1994_T		MinTermFlex	5		NormGrossAnnualPrem	0,04226001	
6	n	30		alpha	2,50%					GrossAnnualPrem	4.226,00 €	
7	t	20		beta1	2,50%					GrossModalPrem	371,88 €	
8	SumInsured	100.000,00 €		gamma1	0,080%							
9	PayFreq	12		gamma2	0,125%					Pxt	0,04001217	
10				gamma3	0,250%							
11				k	24,00							
12				ModalSurcharge	5%							
13												
14	**Progression values**											
15	k	Axn	axn	axt	kVx_pp	kDRx_pp	kVx_pu	kVx_MRV	Flex. phase	Surrender deduction	Surrender value	SumInsured_pu
16	0	0,6315923	21,4202775	16,3130941	-0,0211300	- 2.113,00 €	0,6851430	- €	0	150,00 €	- €	- €
17	1	0,6417247	20,8311476	15,6212042	0,0170857	1.708,57 €	0,6938026	3.415,41 €	0	150,00 €	3.265,41 €	4.922,74 €
18	2	0,6519724	20,2353170	14,9191921	0,0558299	5.582,99 €	0,7025607	6.875,82 €	0	150,00 €	6.725,82 €	9.786,80 €
19	3	0,6623396	19,6325407	14,2066338	0,0951241	9.512,41 €	0,7114209	10.383,03 €	0	150,00 €	10.233,03 €	14.594,78 €
20	4	0,6728224	19,0230412	13,4834183	0,1349695	13.496,95 €	0,7203800	13.936,78 €	0	150,00 €	13.786,78 €	19.346,42 €
21	5	0,6834172	18,4070290	12,7493823	0,1753695	17.536,95 €	0,7294348	17.536,95 €	0	150,00 €	17.386,95 €	24.041,83 €
22	6	0,6941229	17,7845688	12,0042138	0,2163360	21.633,60 €	0,7385843	21.633,60 €	0	150,00 €	21.483,60 €	29.290,63 €
23	7	0,7049358	17,1558776	11,2476494	0,2578765	25.787,65 €	0,7478255	25.787,65 €	0	150,00 €	25.637,65 €	34.483,52 €
24	8	0,7158563	16,5209292	10,4792011	0,3000119	30.001,19 €	0,7571586	30.001,19 €	0	150,00 €	29.851,19 €	39.623,39 €
25	9	0,7268794	15,8800126	9,6985065	0,3427526	34.275,26 €	0,7665794	34.275,26 €	0	150,00 €	34.125,26 €	44.711,94 €
26	10	0,7380032	15,2332442	8,9049974	0,3861203	38.612,03 €	0,7760863	38.612,03 €	0	150,00 €	38.462,03 €	49.752,24 €
27	11	0,7492252	14,5807607	8,0980045	0,4301409	43.014,09 €	0,7856771	43.014,09 €	0	150,00 €	42.864,09 €	54.747,79 €
28	12	0,7605444	13,9226349	7,2766856	0,4748482	47.484,82 €	0,7953509	47.484,82 €	0	150,00 €	47.334,82 €	59.702,97 €
29	13	0,7719607	13,2588561	6,4399924	0,5202860	52.028,60 €	0,8051078	52.028,60 €	0	150,00 €	51.878,60 €	64.623,14 €
30	14	0,7834747	12,5893974	5,5866734	0,5665069	56.650,69 €	0,8149482	56.650,69 €	0	150,00 €	56.500,69 €	69.514,47 €
31	15	0,7950966	11,9136717	4,7150097	0,6135919	61.359,19 €	0,8248807	61.359,19 €	0	150,00 €	61.209,19 €	74.385,53 €
32	16	0,8068404	11,2308508	3,8229774	0,6616385	66.163,85 €	0,8349175	66.163,85 €	0	150,00 €	66.013,85 €	79.245,98 €
33	17	0,8187232	10,5399537	2,9082675	0,7107585	71.075,85 €	0,8450730	71.075,85 €	0	150,00 €	70.925,85 €	84.106,17 €

Calculation | MortalityTables

	A	B	C	D	E
1					
2					
3	x/y	DAV1994_T_M	DAV1994_T_F	DAV2008_T_M	DAV2008_T_F
4	0	0,01168700	0,00900300	0,00611300	0,00508800
5	1	0,00100800	0,00086700	0,00042300	0,00038700
6	2	0,00072800	0,00062400	0,00034300	0,00031800
7	3	0,00054200	0,00044400	0,00027500	0,00025500
8	4	0,00047300	0,00034500	0,00022000	0,00020200
9	5	0,00045200	0,00030700	0,00018200	0,00016300
10	6	0,00043300	0,00029300	0,00015500	0,00013400
11	7	0,00040800	0,00028300	0,00013900	0,00011500
12	8	0,00037900	0,00027500	0,00012900	0,00010500
13	9	0,00035200	0,00026800	0,00012500	0,00009900
14	10	0,00033400	0,00026100	0,00012900	0,00010200
15	11	0,00033100	0,00026000	0,00014300	0,00011100
16	12	0,00034000	0,00026700	0,00017300	0,00012700
17	13	0,00037100	0,00028100	0,00022200	0,00015300
18	14	0,00045100	0,00030700	0,00030300	0,00018800
19	15	0,00059300	0,00035300	0,00041700	0,00022800
20	16	0,00079200	0,00041600	0,00055700	0,00027100
21	17	0,00104000	0,00048000	0,00070900	0,00031000
22	18	0,00129800	0,00053700	0,00085000	0,00032400
23	19	0,00143700	0,00056000	0,00095300	0,00033000
24	20	0,00147600	0,00056000	0,00101200	0,00032800
25	21	0,00147600	0,00056000	0,00102200	0,00032200
26	22	0,00147600	0,00056000	0,00100400	0,00031400
27	23	0,00147600	0,00056000	0,00096300	0,00030400
28	24	0,00147600	0,00056000	0,00091100	0,00029700
29	25	0,00147600	0,00056000	0,00085600	0,00029300
30	26	0,00147600	0,00056000	0,00080800	0,00029200
31	27	0,00147600	0,00058100	0,00077200	0,00029200
32	28	0,00147600	0,00061200	0,00075200	0,00029600
33	29	0,00147600	0,00064500	0,00074500	0,00030200
34	30	0,00147600	0,00068900	0,00075200	0,00031100
35	31	0,00147600	0,00073500	0,00076800	0,00032700
36	32	0,00148900	0,00078300	0,00079100	0,00035100
37	33	0,00155100	0,00083300	0,00082000	0,00038600
38	34	0,00164100	0,00089700	0,00085500	0,00043300
39	35	0,00174700	0,00097100	0,00089500	0,00049000
40	36	0,00186900	0,00105700	0,00094500	0,00055500
41	37	0,00200700	0,00115600	0,00100500	0,00062400

Worksheet for Calculation

Endowment Life Insurance Calculator

VBA solution with caching commutation values

Policy data	
x	40
Sex	M
n	30
t	20
SumInsured	100.000,00 €
PayFreq	12

Tariff data	
InterestRate	1,75%
MortalityTable	DAV1994_T
alpha	2,50%
beta1	2,50%
gamma1	0,080%
gamma2	0,125%
gamma3	0,250%
k	24,00
ModalSurcharge	5%

Limits	
MinAgeFlex	60
MinTermFlex	5

Premium calculation	
NormGrossAnnualPrem	0,04226001
GrossAnnualPrem	4.226,00 €
GrossModalPrem	371,88 €
Pxt	0,04001217

Progression values

k	Axn	axn	axt	kVx_pp	kDRx_pp	kVx_pu	kVx_MRV	Flex. phase	Surrender deduction	Surrender value	SumInsured_pu
0	0,6315923	21,4202775	16,3130941	-0,0211300	- 2.113,00 €	0,6851430	- €	0	150,00 €	- €	- €
1	0,6417247	20,8311476	15,6212042	0,0170857	1.708,57 €	0,6938026	3.415,41 €	0	150,00 €	3.265,41 €	4.922,74 €
2	0,6519724	20,2353170	14,9191921	0,0558299	5.582,99 €	0,7025607	6.875,82 €	0	150,00 €	6.725,82 €	9.786,80 €
3	0,6623396	19,6325407	14,2066338	0,0951241	9.512,41 €	0,7114209	10.383,03 €	0	150,00 €	10.233,03 €	14.594,78 €
4	0,6728224	19,0230412	13,4834183	0,1349695	13.496,95 €	0,7203800	13.936,78 €	0	150,00 €	13.786,78 €	19.346,42 €
5	0,6834172	18,4070290	12,7493823	0,1753695	17.536,95 €	0,7294348	17.536,95 €	0	150,00 €	17.386,95 €	24.041,83 €
6	0,6941229	17,7845688	12,0042138	0,2163360	21.633,60 €	0,7385843	21.633,60 €	0	150,00 €	21.483,60 €	29.290,63 €
7	0,7049358	17,1558776	11,2476494	0,2578765	25.787,65 €	0,7478255	25.787,65 €	0	150,00 €	25.637,65 €	34.483,52 €
8	0,7158563	16,5209292	10,4792011	0,3000119	30.001,19 €	0,7571586	30.001,19 €	0	150,00 €	29.851,19 €	39.623,39 €
9	0,7268794	15,8800126	9,6985065	0,3427526	34.275,26 €	0,7665794	34.275,26 €	0	150,00 €	34.125,26 €	44.711,94 €
10	0,7380032	15,2332442	8,9049974	0,3861203	38.612,03 €	0,7760863	38.612,03 €	0	150,00 €	38.462,03 €	49.752,24 €
11	0,7492252	14,5807607	8,0980045	0,4301409	43.014,09 €	0,7856771	43.014,09 €	0	150,00 €	42.864,09 €	54.747,79 €
12	0,7605444	13,9226349	7,2766856	0,4748482	47.484,82 €	0,7953509	47.484,82 €	0	150,00 €	47.334,82 €	59.702,97 €
13	0,7719607	13,2588561	6,4399924	0,5202860	52.028,60 €	0,8051078	52.028,60 €	0	150,00 €	51.878,60 €	64.623,14 €
14	0,7834747	12,5893974	5,5866734	0,5665069	56.650,69 €	0,8149482	56.650,69 €	0	150,00 €	56.500,69 €	69.514,47 €
15	0,7950966	11,9136717	4,7150097	0,6135919	61.359,19 €	0,8248807	61.359,19 €	0	150,00 €	61.209,19 €	74.385,53 €
16	0,8068404	11,2308508	3,8229774	0,6616385	66.163,85 €	0,8349175	66.163,85 €	0	150,00 €	66.013,85 €	79.245,98 €
17	0,8187232	10,5399537	2,9082675	0,7107585	71.075,85 €	0,8450730	71.075,85 €	0	150,00 €	70.925,85 €	84.106,17 €

Calculation | MortalityTables

Input

Worksheet for Calculation

Endowment Life Insurance Calculator

VBA solution with caching commutation values

Policy data		Tariff data		Limits		Premium calculation	
x	40	InterestRate	1,75%	MinAgeFlex	60		
Sex	M	MortalityTable	DAV1994_T	MinTermFlex	5	NormGrossAnnualPrem	0,04226001
n	30	alpha	2,50%			GrossAnnualPrem	4.226,00 €
t	20	beta1	2,50%			GrossModalPrem	371,88 €
SumInsured	100.000,00 €	gamma1	0,080%				
PayFreq	12	gamma2	0,125%			Pxt	0,04001217
		gamma3	0,250%				
		k	24,00				
		ModalSurcharge	5%				

Progression values

k	Axn	axn	axt	kVx_pp	kDRx_pp	kVx_pu	kVx_MRV	Flex. phase	Surrender deduction	Surrender value	SumInsured_pu
0	0,6315923	21,4202775	16,3130941	-0,0211300	- 2.113,00 €	0,6851430	- €	0	150,00 €	- €	- €
1	0,6417247	20,8311476	15,6212042	0,0170857	1.708,57 €	0,6938026	3.415,41 €	0	150,00 €	3.265,41 €	4.922,74 €
2	0,6519724	20,2353170	14,9191921	0,0558299	5.582,99 €	0,7025607	6.875,82 €	0	150,00 €	6.725,82 €	9.786,80 €
3	0,6623396	19,6325407	14,2066338	0,0951241	9.512,41 €	0,7114209	10.383,03 €	0	150,00 €	10.233,03 €	14.594,78 €
4	0,6728224	19,0230412	13,4834183	0,1349695	13.496,95 €	0,7203800	13.936,78 €	0	150,00 €	13.786,78 €	19.346,42 €
5	0,6834172	18,4070290	12,7493823	0,1753695	17.536,95 €	0,7294348	17.536,95 €	0	150,00 €	17.386,95 €	24.041,83 €
6	0,6941229	17,7845688	12,0042138	0,2163360	21.633,60 €	0,7385843	21.633,60 €	0	150,00 €	21.483,60 €	29.290,63 €
7	0,7049358	17,1558776	11,2476494	0,2578765	25.787,65 €	0,7478255	25.787,65 €	0	150,00 €	25.637,65 €	34.483,52 €
8	0,7158563	16,5209292	10,4792011	0,3000119	30.001,19 €	0,7571586	30.001,19 €	0	150,00 €	29.851,19 €	39.623,39 €
9	0,7268794	15,8800126	9,6985065	0,3427526	34.275,26 €	0,7665794	34.275,26 €	0	150,00 €	34.125,26 €	44.711,94 €
10	0,7380032	15,2332442	8,9049974	0,3861203	38.612,03 €	0,7760863	38.612,03 €	0	150,00 €	38.462,03 €	49.752,24 €
11	0,7492252	14,5807607	8,0980045	0,4301409	43.014,09 €	0,7856771	43.014,09 €	0	150,00 €	42.864,09 €	54.747,79 €
12	0,7605444	13,9226349	7,2766856	0,4748482	47.484,82 €	0,7953509	47.484,82 €	0	150,00 €	47.334,82 €	59.702,97 €
13	0,7719607	13,2588561	6,4399924	0,5202860	52.028,60 €	0,8051078	52.028,60 €	0	150,00 €	51.878,60 €	64.623,14 €
14	0,7834747	12,5893974	5,5866734	0,5665069	56.650,69 €	0,8149482	56.650,69 €	0	150,00 €	56.500,69 €	69.514,47 €
15	0,7950966	11,9136717	4,7150097	0,6135919	61.359,19 €	0,8248807	61.359,19 €	0	150,00 €	61.209,19 €	74.385,53 €
16	0,8068404	11,2308508	3,8229774	0,6616385	66.163,85 €	0,8349175	66.163,85 €	0	150,00 €	66.013,85 €	79.245,98 €
17	0,8187232	10,5399537	2,9082675	0,7107585	71.075,85 €	0,8450730	71.075,85 €	0	150,00 €	70.925,85 €	84.106,17 €

Calculation | MortalityTables

Output

Worksheet for Calculation

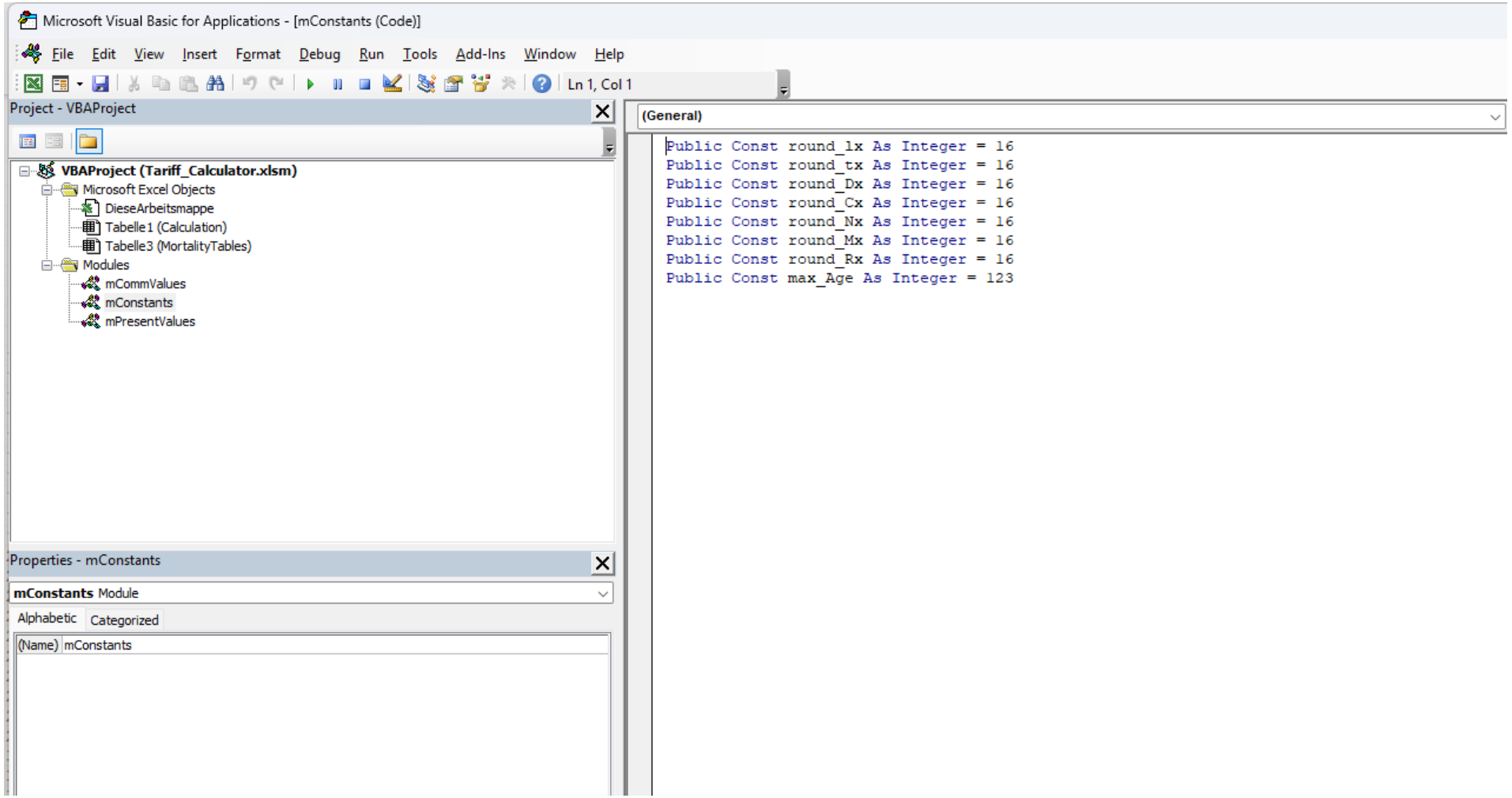
	A	B	C	D	E	F	G	H	I	J	K	L
1	**Endowment Life Insurance Calculator**									VBA solution with caching commutation values		
2												
3	**Policy data**			**Tariff data**			**Limits**			**Premium calculation**		
4	x	40		InterestRate	1,75%		MinAgeFlex	60				
5	Sex	M		MortalityTable	DAV1994_T		MinTermFlex	5		NormGrossAnnualPrem	0,04226001	
6	n	30		alpha	2,50%					GrossAnnualPrem	4.226,00 €	
7	t	20		beta1	2,50%					GrossModalPrem	371,88 €	
8	SumInsured	100.000,00 €		gamma1	0,080%							
9	PayFreq	12		gamma2	0,125%					Pxt	0,04001217	
10				gamma3	0,250%							
11				k	24,00							
12				ModalSurcharge	5%							
13												
14	**Progression values**											
15	k	Axn	axn	axt	kVx_pp	kDRx_pp	kVx_pu	kVx_MRV	Flex. phase	Surrender deduction	Surrender value	SumInsured_pu
16	0	0,6315923	21,4202775	16,3130941	-0,0211300	- 2.113,00 €	0,6851430	- €	0	150,00 €	- €	- €
17	1	0,6417247	20,8311476	15,6212042	0,0170857	1.708,57 €	0,6938026	3.415,41 €	0	150,00 €	3.265,41 €	4.922,74 €
18	2	0,6519724	20,2353170	14,9191921	0,0558299	5.582,99 €	0,7025607	6.875,82 €	0	150,00 €	6.725,82 €	9.786,80 €
19	3	0,6623396	19,6325407	14,2066338	0,0951241	9.512,41 €	0,7114209	10.383,03 €	0	150,00 €	10.233,03 €	14.594,78 €
20	4	0,6728224	19,0230412	13,4834183	0,1349695	13.496,95 €	0,7203800	13.936,78 €	0	150,00 €	13.786,78 €	19.346,42 €
21	5	0,6834172	18,4070290	12,7493823	0,1753695	17.536,95 €	0,7294348	17.536,95 €	0	150,00 €	17.386,95 €	24.041,83 €
22	6	0,6941229	17,7845688	12,0042138	0,2163360	21.633,60 €	0,7385843	21.633,60 €	0	150,00 €	21.483,60 €	29.290,63 €
23	7	0,7049358	17,1558776	11,2476494	0,2578765	25.787,65 €	0,7478255	25.787,65 €	0	150,00 €	25.637,65 €	34.483,52 €
24	8	0,7158563	16,5209292	10,4792011	0,3000119	30.001,19 €	0,7571586	30.001,19 €	0	150,00 €	29.851,19 €	39.623,39 €
25	9	0,7268794	15,8800126	9,6985065	0,3427526	34.275,26 €	0,7665794	34.275,26 €	0	150,00 €	34.125,26 €	44.711,94 €
26	10	0,7380032	15,2332442	8,9049974	0,3861203	38.612,03 €	0,7760863	38.612,03 €	0	150,00 €	38.462,03 €	49.752,24 €
27	11	0,7492252	14,5807607	8,0980045	0,4301409	43.014,09 €	0,7856771	43.014,09 €	0	150,00 €	42.864,09 €	54.747,79 €
28	12	0,7605444	13,9226349	7,2766856	0,4748482	47.484,82 €	0,7953509	47.484,82 €	0	150,00 €	47.334,82 €	59.702,97 €
29	13	0,7719607	13,2588561	6,4399924	0,5202860	52.028,60 €	0,8051078	52.028,60 €	0	150,00 €	51.878,60 €	64.623,14 €
30	14	0,7834747	12,5893974	5,5866734	0,5665069	56.650,69 €	0,8149482	56.650,69 €	0	150,00 €	56.500,69 €	69.514,47 €
31	15	0,7950966	11,9136717	4,7150097	0,6135919	61.359,19 €	0,8248807	61.359,19 €	0	150,00 €	61.209,19 €	74.385,53 €
32	16	0,8068404	11,2308508	3,8229774	0,6616385	66.163,85 €	0,8349175	66.163,85 €	0	150,00 €	66.013,85 €	79.245,98 €
33	17	0,8187232	10,5399537	2,9082675	0,7107585	71.075,85 €	0,8450730	71.075,85 €	0	150,00 €	70.925,85 €	84.106,17 €

Calculation | MortalityTables

=(act_nGrAx(x;n;Sex;MortalityTable;InterestRate) + Act_Dx(x+n;Sex;MortalityTable;InterestRate) / Act_Dx(x;Sex;MortalityTable;InterestRate) + gamma1 * Act_axn_k(x;t;Sex;MortalityTable;InterestRate;1) + gamma2 * (Act_axn_k(x;n;Sex;MortalityTable;InterestRate;1) - Act_axn_k(x;t;Sex;MortalityTable;InterestRate;1))) / ((1-beta1) * Act_axn_k(x;t;Sex;MortalityTable;InterestRate;1) - alpha*t)

=IF(A16<=n; act_nGrAx(x+$A16;MAX(0;n-$A16); Sex; MortalityTable; InterestRate)+Act_Dx(x+n; Sex; MortalityTable; InterestRate) / Act_Dx(x+$A16; Sex; MortalityTable; InterestRate); 0)

VBA Module "mConstants"



VBA Module "mCommValues"

```
Dim cache As Object
```

```
Public Sub InitializeCache()
    ' Create a new Dictionary object
    Set cache = CreateObject("Scripting.Dictionary")
End Sub
```

```
Public Function Act_qx(Age As Integer, Sex As String, TableId As String, Optional BirthYear As Integer, Optional RetirementAge As Integer, Optional Layer As Integer = 1) As Double
    Dim ws As Worksheet
    Dim tableVector As String

    Set ws = ThisWorkbook.Worksheets("MortalityTables")
    If UCase(Sex) <> "M" Then Sex = "F"

    Select Case UCase(TableId)     ' Check whether the table string is implemented at all
        ' The complete list of all implemented tables must be specified here
        Case "DAV1994_T", "DAV2008_T"
            tableVector = UCase(TableId) & "_" & Sex
            Act_qx = WorksheetFunction.Index(ws.Range("m_Tables"), Age + 1, WorksheetFunction.Match(tableVector, ws.Range("v_Tables"), 0))
        Case Else
            Act_qx = 1#
            Error (1)
    End Select
End Function
```

```
Private Function Vec_lx(EndAge As Integer, Sex As String, TableId As String, Optional BirthYear As Integer, Optional RetirementAge As Integer, Optional Layer As Integer = 1) As Varia
    ' Creates vector of lx
    ' If EndAge = -1 then it is created up to max_Age
    Dim vec() As Variant
    Dim i As Integer
    Dim limit As Integer

    If EndAge = -1 Then
```

VBA Module "mPresentValues"

```
Public Function Act_ax_k(Age As Integer, Sex As String, TableId As String, InterestRate As Double, k As Integer, Optional BirthYear As Integer, Optional RetirementAge As Integer, Optic
    If k > 0 Then
        Act_ax_k = Act_Nx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) / _
                   Act_Dx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) - _
                   Act_DeductionTerm(k, InterestRate)
    Else
        Act_ax_k = 0
    End If
End Function
```

```
Public Function Act_axn_k(Age As Integer, n As Integer, Sex As String, TableId As String, InterestRate As Double, k As Integer, Optional BirthYear As Integer, Optional RetirementAge As
    If k > 0 Then
        Act_axn_k = (Act_Nx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) - _
                     Act_Nx(Age + n, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer)) / _
                    Act_Dx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) - _
                    Act_DeductionTerm(k, InterestRate) * (1 - _
                    Act_Dx(Age + n, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) / _
                    Act_Dx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer))
    Else
        Act_axn_k = 0
    End If
End Function
```

```
Public Function Act_nax_k(Age As Integer, n As Integer, Sex As String, TableId As String, InterestRate As Double, k As Integer, Optional BirthYear As Integer, Optional RetirementAge As
    If k > 0 Then
        Act_nax_k = Act_Dx(Age + n, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) / _
                    Act_Dx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) * _
                    Act_ax_k(Age + n, Sex, TableId, InterestRate, k, BirthYear, RetirementAge, Layer)
    Else
        Act_nax_k = 0
    End If
End Function
```

```
Public Function act_nGrAx(Age As Integer, n As Integer, Sex As String, TableId As String, InterestRate As Double, Optional BirthYear As Integer, Optional RetirementAge As Integer, Opti
    act_nGrAx = (Act_Mx(Age, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer) - _
                 Act_Mx(Age + n, Sex, TableId, InterestRate, BirthYear, RetirementAge, Layer)) / _
```